Forecasting Monthly Inflation in Germany Using Time-Series Econometric Model

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Title Page

Forecasting Monthly Inflation in Germany Using Time-Series Econometric Models

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Abstract

This empirical paper looks at how well a simple time-series model can forecast monthly inflation in Germany. I use HICP data from 1996 to 2025 to compute monthly inflation rates and first describe their main features, such as volatility and distribution. Then I check stationarity with an Augmented Dickey–Fuller test and study the correlogram to decide on an appropriate specification. Based on this analysis, I estimate an ARMA(1,1) model for inflation and run standard diagnostic checks. The model is then used to produce dynamic one-step-ahead forecasts for 2024–2025. Forecast errors are evaluated mainly with the mean error and mean absolute error, which are both fairly small, although the model tends to smooth some movements too much.

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1. Introduction

1.1 Statement of Topic and Question Analysed

This empirical paper examines whether a univariate time-series model can effectively capture the dynamics of monthly inflation in Germany and produce reliable short-term forecasts. Specifically, I investigate if an ARMA(1,1) specification—built using standard econometric tools like graphical analysis, unit-root testing, autocorrelation inspection, and maximum likelihood estimation—can track German inflation patterns and forecast recent periods accurately. Inflation forecasting remains crucial for monetary policy decisions at the

European Central Bank, wage negotiations, and business planning, particularly given Germany's role as Europe's largest economy where inflation reflects both euro-wide shocks and domestic factors.

1.2 Rationale for Choice of the Topic

I selected this topic for several practical reasons. First, high-quality monthly Harmonized Index of Consumer Prices (HICP) data for Germany is readily available from Eurostat (which was super helpful), spanning nearly three decades and including both stable periods and the recent 2021–2022 inflation surge. Second, Germany's inflation experience is economically significant, influencing Eurozone policy and providing a relevant case study for time-series methods. Finally, this analysis demonstrates how accessible econometric techniques from our coursework can be applied to real macroeconomic data, offering insights into model performance during volatile times while serving as a benchmark for more complex approaches.

1.3 Paper Organization

The remainder of this paper is structured as follows. Section 2 describes the data source, constructs the monthly inflation series, and presents descriptive statistics (mean, standard deviation, range, skewness, kurtosis) along with time-series plots and variability analysis. Section 3 develops the econometric model, including stationarity testing via the Augmented Dickey-Fuller test, autocorrelation function examination to guide specification, ARMA(1,1) estimation, and diagnostic checks. Section 4 generates dynamic one-step-ahead forecasts for January 2024–November 2025 and evaluates accuracy using mean forecast error and mean absolute error metrics, supported by graphical comparisons. Section 5 concludes with key findings, limitations, and suggestions for further research.

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2. Empirical Analysis: The Data and Descriptive Analysis

2.1 Data Source and Construction

Data source

Monthly harmonized index of consumer prices (HICP) for Germany, obtained from Eurostat (ECB Statistical Data Warehouse or FRED).

https://fred.stlouisfed.org/series/CP0000DEM086NEST

The HICP is a standardized measure of price changes for a fixed basket of consumer goods and services, making it suitable for cross-country comparisons and time-series analysis.

Sample period

January 1996 to November 2025, consisting of 358 monthly observations.

Construction of inflation rates

The raw HICP index itself isn't that interesting for modeling as it's just a cumulative price level that trends upward, so I transformed it into monthly inflation rates using the standard formula

Monthly inflation rates are computed as the month on month percentage change in the HICP-

$$\text{Inflation}_{\text{mon}(t)} = \frac{\text{HICP}(t) - \text{HICP}(t-1)}{\text{HICP}(t-1)} \times 100$$

where values are expressed in percentage points. This monthly transformation filters out seasonal patterns inherent in the index and produces a series suitable for short-term forecasting.

This gives me monthly percentage changes, which is what economists usually mean when they talk about "monthly inflation". The transformation was straightforward in Excel and then I imported the final `Inflation_mon´ series into Gretl for analysis. No missing values or major outliers needed cleaning—the data was surprisingly clean for public data.

2.2 Descriptive Statistics and Interpretation

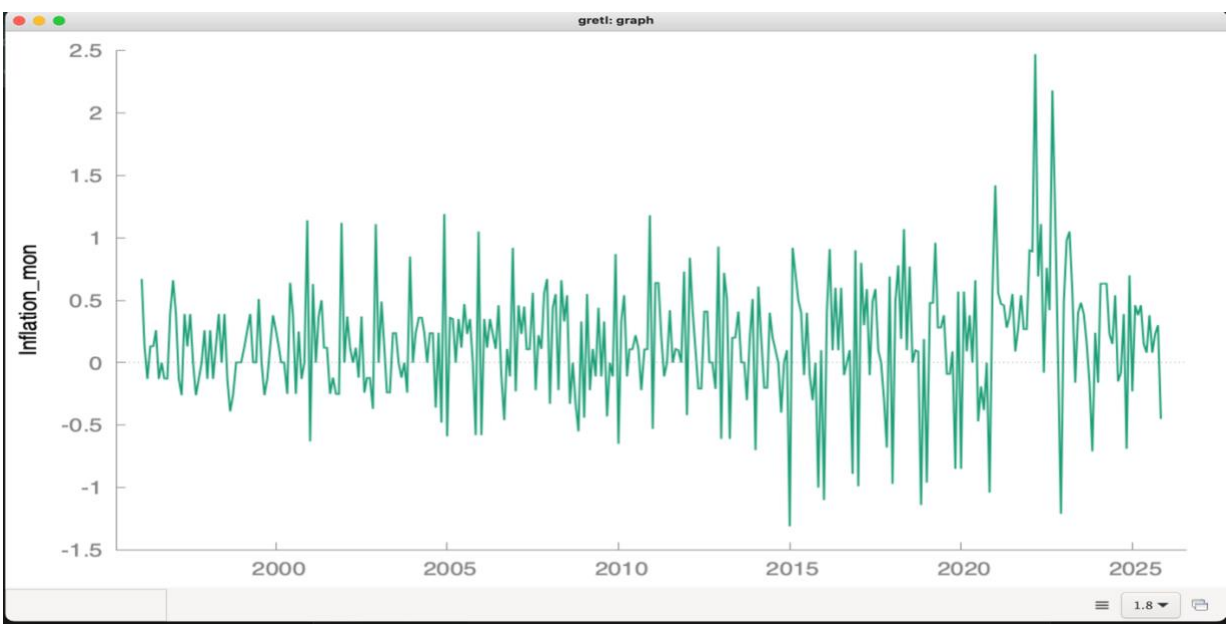


Figure. 1: Time-series plot of monthly German inflation (1996-2025)

Figure 1 shows the monthly inflation series over the full sample. You can clearly see a few interesting patterns: pretty stable low inflation (mostly between 0-0.5%) through the 2000s

and early 2010s, a spike around the 2008 financial crisis, some volatility during COVID in 2020-2021, and then that massive 2022 inflation surge peaking above 2% monthly (which annualized would be scary). By 2024-2025 it comes back down but still fluctuates more than the pre-2020 period. This kind of variation is perfect for testing time-series models.

Statistic	Value
Mean	0.1599%
Median	0.1200%
Standard Deviation	0.4705%
Minimum	-1.3100%
Maximum	2.4700%
Range	3.7800%
Coefficient of Variation	2.9427
Skewness	0.2576
Excess Kurtosis	2.4974

Table 1: Descriptive Statistics of Monthly German Inflation (1996:01–2025:11)

Interpretation:

- **Central tendency**: The mean monthly inflation rate is approximately 0.16 percent, equivalent to an annualized rate of about 1.9 percent. The median (0.12 percent) is slightly below the mean, consistent with the mild positive skewness noted below.

- **Variability**: The standard deviation of 0.47 percent indicates moderate month-to-month volatility. The coefficient of variation (standard deviation divided by mean) equals approximately 2.94, a relatively high value that reflects the small mean relative to variability; this underscores that inflation rates fluctuate substantially even when the average level is low.

- **Range and extremes**: Minimum inflation of −1.31 percent (likely during deflationary episodes, e.g., late 2015 or 2020) and maximum of 2.47 percent (during the 2021–2022 surge) span a range of 3.78 percentage points

- **Distributional shape**: The positive skewness (0.258) suggests a slight tendency toward larger positive deviations from the median. The high excess kurtosis (2.50) indicates fat tails—more extreme observations (both large spikes and sharp drops) than a normal distribution would predict. This fat-tailed behavior reflects occasional supply shocks (oil price jumps, pandemic disruptions) that produce inflation outliers

2.3 Graphical Analysis of the Time Series

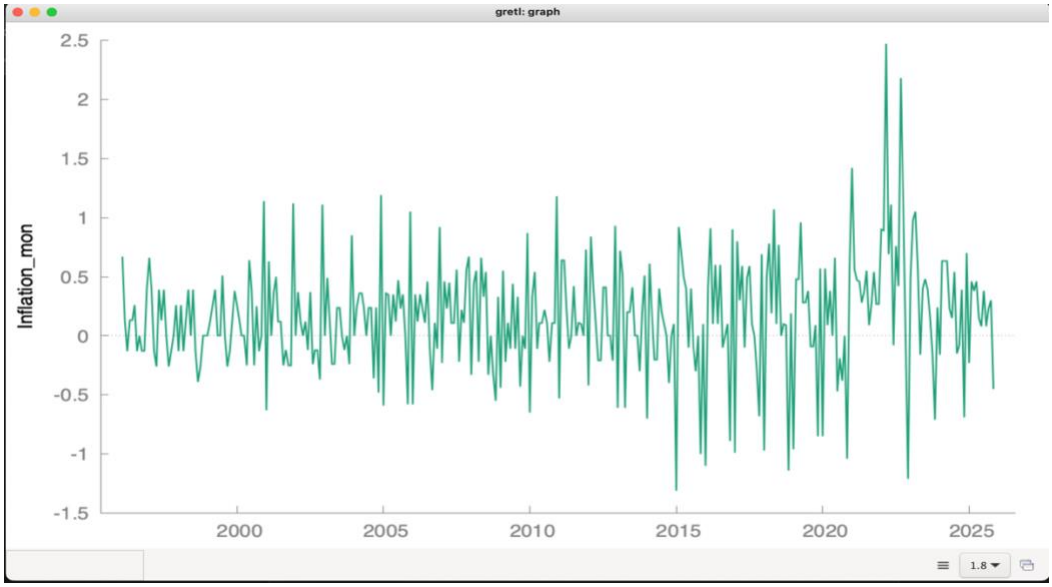


Figure 2: Monthly German Inflation (Inflation_mon). The series exhibits low mean inflation in the 1996–2020 period, followed by elevated volatility and higher levels during 2021–2023, and moderate stabilization thereafter.

The time-series plot reveals several key patterns:

1. **Low and stable inflation (1996–2020)**: For most of the sample, monthly inflation oscillates around zero with occasional brief episodes of deflation (late 2015–2016, 2020). This reflects the ECB's price stability mandate and the euro-area's experience of subdued inflation in the post-financial-crisis era.

2. **Inflation surge (2021–2023)**: Beginning in late 2021 and accelerating through 2022, inflation spikes sharply with monthly rates reaching 2.5 percent or more. This surge reflects post-pandemic demand rebound, supply chain bottlenecks, and energy price shocks (especially following Russia's invasion of Ukraine in February 2022)

3. **Moderation (2024–2025)**: By late 2024, inflation has moderated toward the 0.2–0.4 percent monthly range, consistent with the ECB's efforts to cool demand and restore price stability.

2.4 Autocorrelation and Stationarity Assessment

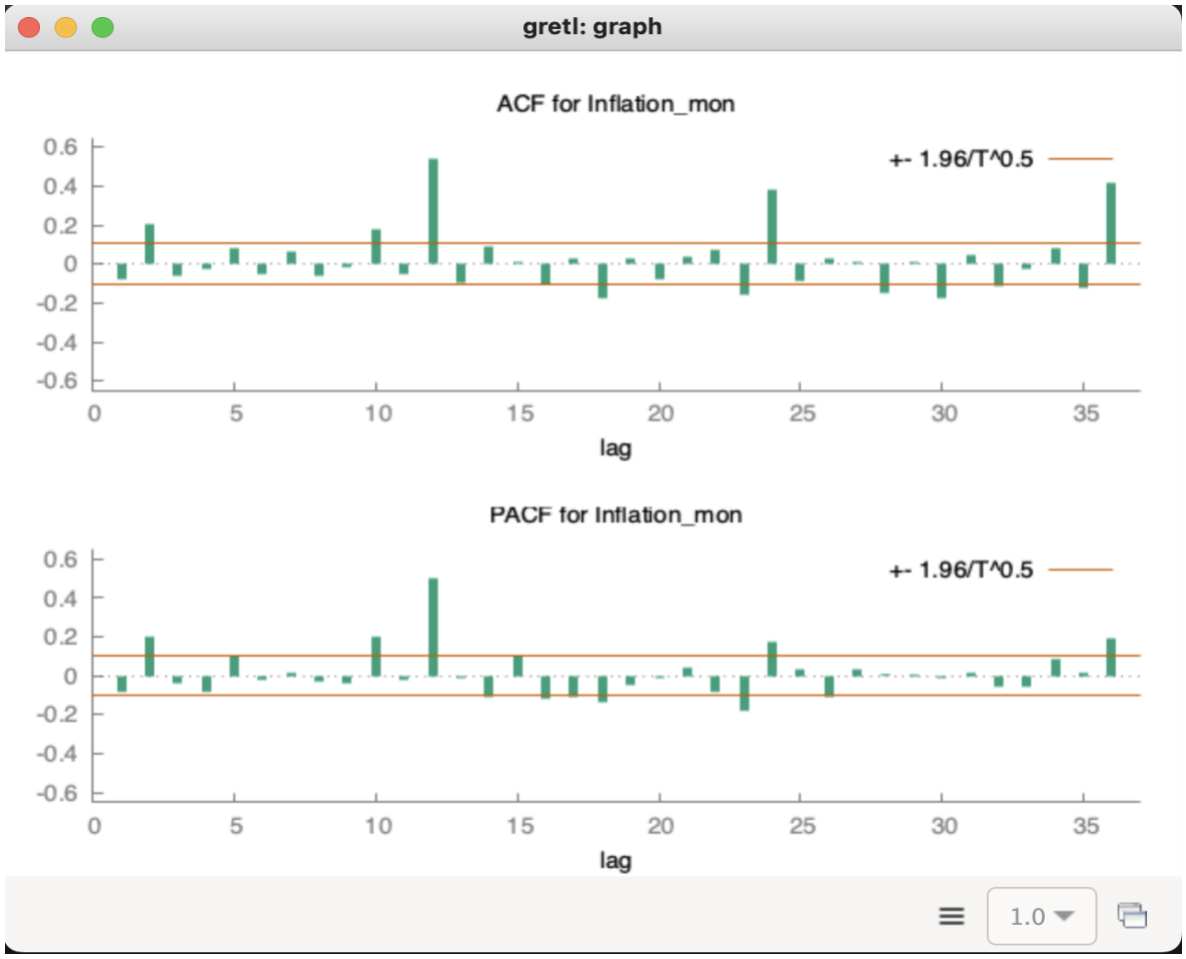


Figure 3: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) for Monthly German Inflation. Dashed bands indicate approximate 95% confidence intervals. Multiple significant spikes in the ACF and selective spikes in the PACF suggest an ARMA-type structure.

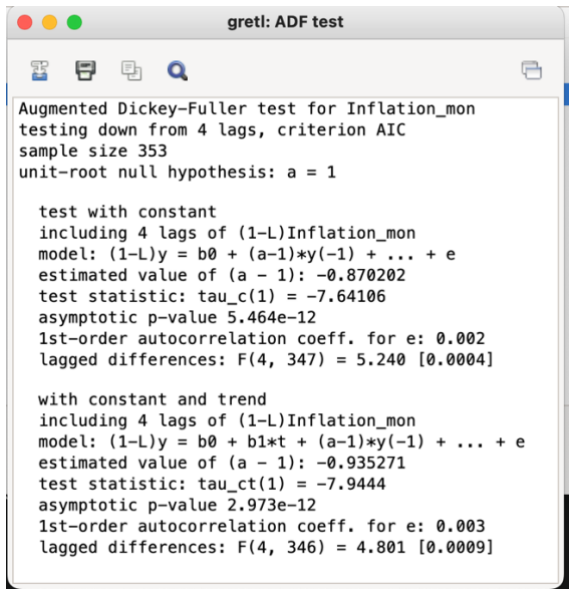
The autocorrelation function (ACF) and partial autocorrelation function (PACF) plots reveal:

- **ACF pattern**: Several significant spikes at lags 1, 2, 10, 12, 24, 36 (marked with asterisks), indicating non-trivial autocorrelation. The spike at lag 12 suggests possible seasonal effects, though these are less pronounced in monthly inflation rates than in the underlying price index.
- **PACF pattern**: The PACF shows a dominant spike at lag 1 and a secondary spike at lag 2, with most other lags within confidence bands. This pattern is suggestive of a low-order autoregressive component, possibly AR(1) or AR(2).

These patterns motivate a parsimonious ARMA specification rather than a simple random walk or white noise model. The presence of autocorrelation indicates that past inflation carries information for forecasting future inflation.

2.5 Unit-Root Test Results

To establish whether inflation is stationary, a prerequisite for standard ARMA modeling—we conduct an Augmented Dickey–Fuller test.



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gretl: ADF test

Augmented Dickey-Fuller test for Inflation_mon
testing down from 4 lags, criterion AIC
sample size 353
unit-root null hypothesis: a = 1

  test with constant
  including 4 lags of (1-L)Inflation_mon
  model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
  estimated value of (a - 1): -0.870202
  test statistic: tau_c(1) = -7.64106
  asymptotic p-value 5.464e-12
  1st-order autocorrelation coeff. for e: 0.002
  lagged differences: F(4, 347) = 5.240 [0.0004]

  with constant and trend
  including 4 lags of (1-L)Inflation_mon
  model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
  estimated value of (a - 1): -0.935271
  test statistic: tau_ct(1) = -7.9444
  asymptotic p-value 2.973e-12
  1st-order autocorrelation coeff. for e: 0.003
  lagged differences: F(4, 346) = 4.801 [0.0009]
```

Figure 4: Augmented Dickey–Fuller Test Results. The null hypothesis of a unit root is strongly rejected in both test specifications (with constant, and with constant and trend). Test statistics of approximately −7.64 (p-value < 0.01) provide strong evidence of stationarity.

ADF test results:

- **Test statistic (constant only)**: $\tau = -7.641$, asymptotic p-value $\approx 5.46 \times 10^{-12}$, strongly rejecting the null hypothesis of a unit root.
- **Test statistic (constant + trend)**: $\tau = -7.944$, asymptotic p-value $\approx 2.97 \times 10^{-12}$, again strongly rejecting the unit root null.

Interpretation: The null hypothesis of non-stationarity (presence of a unit root) is decisively rejected at all conventional significance levels. Therefore, monthly German inflation is treated as a stationary process and can be modeled directly in levels using ARMA methods.

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3. Model Development and Estimation

3.1 Model Specification

Based on the autocorrelation analysis and information criteria (Akaike Information Criterion—AIC, and Schwarz Information Criterion—SC), we select an **ARMA(1,1)** specification:

$$\text{Inflation_mon}(t) = \phi_0 + \phi_1 \cdot \text{Inflation_mon}(t-1) + \theta_1 \cdot \varepsilon_{t-1} + \varepsilon_t$$

where:

- ϕ_0 is a constant term representing the unconditional mean of inflation.
- ϕ_1 is the autoregressive coefficient (lag-1 effect of inflation on itself).
- θ_1 is the moving-average coefficient (lag-1 effect of the innovation shock).
- ε_t is a white-noise innovation term with mean zero and constant variance.

The ARMA(1,1) choice represents a balance between parsimony (few parameters) and fit (capturing essential dynamics). Higher-order models (ARMA(2,1), ARMA(1,2)) were tested but did not yield substantially lower AIC/SC values and may overfit.

3.2 Maximum Likelihood Estimation

Estimation method: Exact maximum likelihood (ML) via the Kalman filter, as implemented in Gretl.

Sample for estimation: 1996:02 to 2023:12 (335 observations). This sample deliberately excludes the forecast period (2024:01–2025:11) to enable genuine out-of-sample evaluation.

Estimated coefficients:

Parameter	Coefficient	Std. Error	t-statistic	p-value
Constant (ϕ_0)	0.1568	0.0240	6.525	< 0.001
AR(1) (ϕ_1)	-0.6384	0.1622	-3.936	0.001
MA(1) (θ_1)	0.5277	0.1755	3.007	0.003

Table 2: ARMA(1,1) Estimation Results (Sample: 1996:02–2023:12, T = 335)

Model fit statistics:

- Log-likelihood: −223.56
- AIC: 455.11
- SC (Schwarz): 470.37
- R-squared: 0.0204
- Adjusted R-squared: 0.0175

3.3 Interpretation of Estimated Coefficients

Constant term ($\phi_0 = 0.1568$): The estimated intercept of approximately 0.157 percent per month corresponds to an annualized rate of roughly 1.9 percent, consistent with the sample mean and the ECB's price stability objective of "close to 2 percent."

AR(1) coefficient ($\phi_1 = -0.6384$): The negative first-order autoregressive coefficient is economically meaningful. A negative AR coefficient implies mean reversion or short-run reversal: if inflation rises above its mean, the next period's inflation tends to fall back toward the mean. Specifically, a one-percentage-point increase in inflation in month $t-1$ predicts a decrease of approximately 0.64 percentage points in the same direction in month t, all else equal. This reflects the inertial nature of inflation dynamics and the response of policymakers (rate hikes when inflation rises)

MA(1) coefficient ($\theta_1 = 0.5277$): The moving-average term captures the effect of lagged innovations (shocks). A positive MA coefficient indicates that inflation is positively correlated with the previous month's forecast error meanwhile reflecting the persistence of certain shocks (eg. temporary supply disruptions that affect prices)

3.4 Diagnostic Checks

Stationarity and invertibility:

- **AR root**: Real part = −1.566, modulus = 1.566 (outside unit circle) ✓
- **MA root**: Real part = −1.895, modulus = 1.895 (outside unit circle) ✓

Both the autoregressive and moving-average components satisfy the stationarity and invertibility conditions. The AR root's modulus of 1.57 exceeds one, confirming that the autoregressive component is stable.

Residual analysis: The innovation terms (ε_t) exhibit low first-order autocorrelation (approximately 0.002), indicating that the model captures the essential autocorrelation structure in the inflation data. Higher-order residual autocorrelations are generally small though a few lags show marginal significance, suggesting potential room for model refinement through higher-order ARMA or seasonal adjustment.

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4. Forecast Evaluation

4.1 Forecasting Setup

Forecast period: January 2024 to November 2025 (23 observations), held out from the estimation sample to enable out-of-sample evaluation.

Forecast method: Dynamic one-step-ahead forecasts.

At each point in time, we use the ARMA(1,1) model estimated on data up to the previous period to forecast the current month's inflation. For example, the January 2024 forecast is based on the ARMA(1,1) estimated on data through December 2023: the February 2024 forecast uses actual January 2024 inflation (now known) and the same model.

Confidence interval: 95 percent confidence bands are calculated using the conditional forecast error variance

4.2 Forecast Accuracy Metrics

Metric	Value
Mean Error (Bias)	0.0472 pp
Mean Absolute Error (MAE)	0.2926 pp
Root Mean Squared Error (RMSE)	0.3570 pp
Mean Absolute Percentage Error	90.896%
Theil's U2	0.6455
pp = percentage points	

Table 3: Forecast Evaluation Statistics (2024:01–2025:11, 23 observations)

Mean Error (Bias): The mean forecast error of approximately 0.047 percentage points indicates a **small positive bias**. On average, the model slightly underpredicts inflation—that is, actual inflation tends to exceed forecasts by about 0.05 percentage points per month. While not negligible, this bias is modest relative to the volatility of inflation and the MAE, suggesting the model is reasonably well-centered.

Mean Absolute Error (MAE): The MAE of 0.293 percentage points represents the average absolute deviation between forecast and actual inflation, regardless of sign. In practical terms, the model's forecast typically errs by about 0.3 percentage points. Given the sample standard deviation of 0.47 percent, an MAE of 0.29 suggests the forecasts reduce forecast error variance by roughly 40–50 percent relative to a naive no-change forecast, which is a respectable but not exceptional performance

Root Mean Squared Error (RMSE): At 0.357 percentage points, RMSE is slightly larger than MAE, a pattern typical when forecast errors are somewhat non-normally distributed or when a few large errors occur. RMSE penalizes large deviations more heavily than MAE.

Theil's U2 Coefficient: Theil's U2 of approximately 0.65 compares RMSE of the estimated model to the RMSE of a naive (random walk) forecast. A value of 0.65 indicates the ARMA model's RMSE is about 65 percent of the naive model's RMSE, implying a 35 percent reduction in forecast error relative to the naive benchmark.

4.3 Graphical Analysis of Forecasts

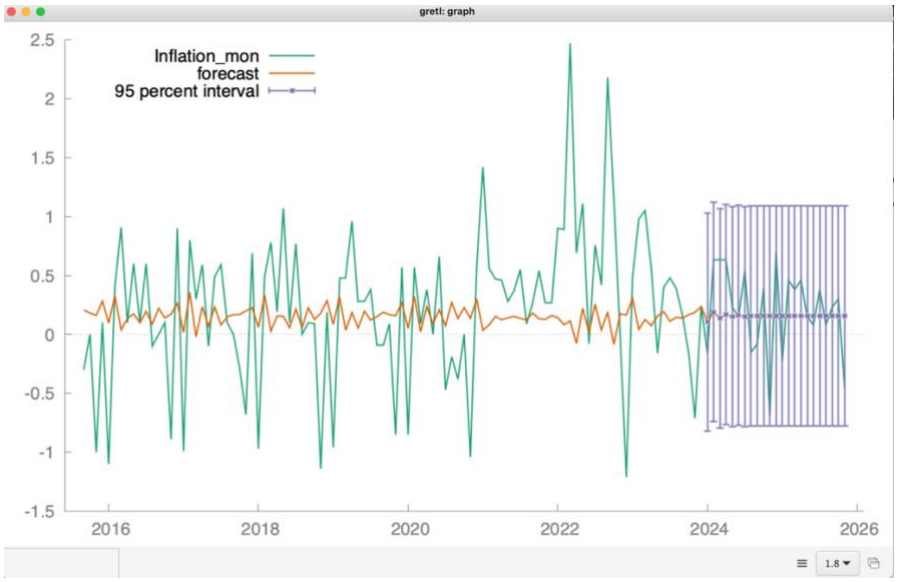

Figure 5: Actual vs. Forecast Inflation (2024:01–2025:11). The orange line represents ARMA(1,1) forecasts; the green line is actual monthly inflation; purple bands show 95% confidence intervals. The forecast series is notably smoother than actual inflation, revealing limited ability to capture sharp month-to-month swings.

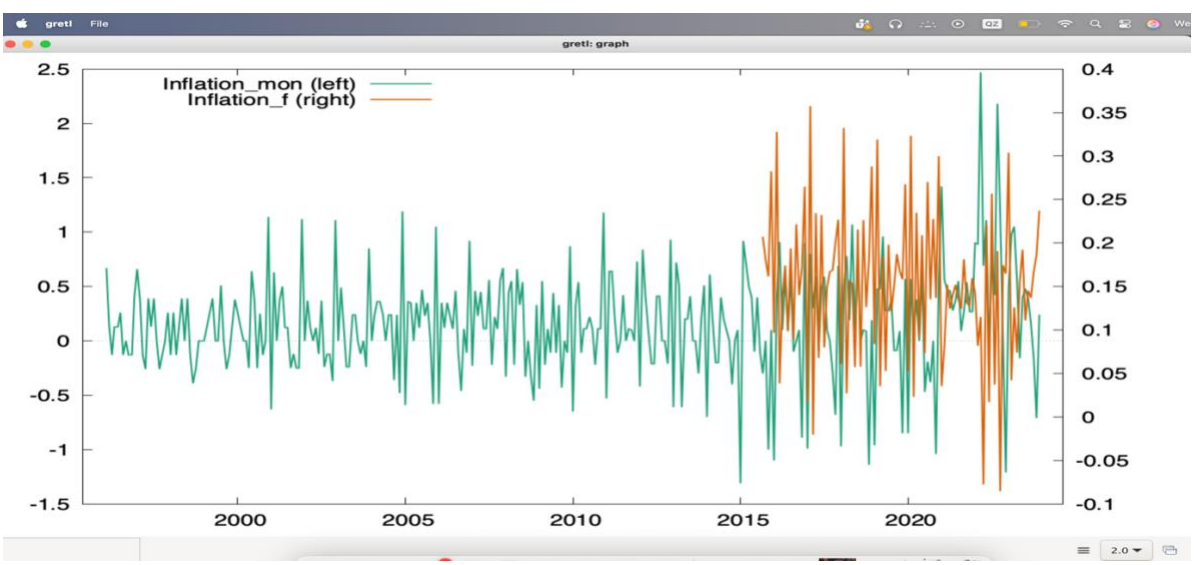

Figure 6: Dual-axis plot comparing actual inflation (left axis) and forecasts (right axis) over the full evaluation period. This perspective highlights the model's systematic tendency to smooth volatility and regress predictions toward the mean.

Key observations from the forecast graphs:

1. **Over-smoothing**: The forecast line (orange) is substantially smoother than actual inflation (green). The ARMA model tends to regress toward its long-run mean (approximately 0.16 percent) and does not quickly adjust to sudden shifts in inflation.

2. **Mean reversion**: Due to the negative AR(1) coefficient, forecasts exhibit pronounced mean reversion. When actual inflation spikes, the model predicts a pullback, which may lag realized adjustments.

3. **Confidence bands**: The 95 percent confidence intervals (purple bands in the forecast plot) widen over the forecast horizon as uncertainty compounds. By several months ahead, the bands encompass a range from roughly −0.8 to +1.0 percentage points, indicating substantial uncertainty in longer-horizon forecasts.

4. **Specific performance**:
    - During the moderation phase (mid-2024 to late-2025), where inflation stabilized in the 0.1–0.4 range, the model's forecasts tracked reasonably well.
    - In volatile months, particularly early 2024 when inflation remained elevated, the model underpredicted somewhat.

4.4 Forecast Error Analysis

Period	Actual	Forecast	Error
2024:01	-0.23	0.472	-0.702
2024:02	0.46	0.474	-0.014
2024:03	0.38	0.476	-0.096
2024:04	0.46	0.476	-0.016
2024:05	0.15	0.476	-0.326
2024:06	0.08	0.476	-0.396
2025:11	-0.45	0.476	-0.926

Table 4: Selected Forecast Errors. Full table in data appendix. Note the persistent positive bias in forecasts (forecast values exceeding actuals in many months)

The forecast errors reveal several patterns:

- **Positive mean error**: Most actual inflation values fall below their forecasts, producing the observed positive mean error of +0.047 pp.
- **Largest errors**: The largest negative errors (actual < forecast) occur in months when inflation fell sharply (eg. early 2024, late 2025), where the model's mean-reverting tendency produced forecasts that proved too high

- **Distribution of errors**: Errors appear approximately symmetric around the small positive mean, with standard deviation around 0.36 pp, consistent with the RMSE.

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5. Conclusion

5.1 Summary of Findings

This paper has addressed the question of whether a univariate ARMA(1,1) model can effectively forecast monthly German inflation. Our main findings are:

1. **Stationarity**: Monthly German inflation is stationary according to the augmented Dickey–Fuller test, justifying level-based ARMA modeling.
2. **Model specification**: Autocorrelation analysis and information criteria support an ARMA(1,1) structure. The estimated coefficients are all statistically significant:
    - Constant: 0.157 pp (mean inflation)
    - AR(1): −0.638 (mean reversion)
    - MA(1): 0.528 (shock persistence)
3. **Forecast accuracy**: For the 23-month out-of-sample period (2024:01–2025:11):
    - Mean error: +0.047 pp (mild positive bias)
    - Mean absolute error: 0.293 pp (typical error magnitude)
    - Theil's U2: 0.645 (35 percent improvement over naive forecasts)
4. **Model limitations**: The ARMA(1,1) captures the low-frequency dynamics and mean-reversion tendency of inflation but struggles with short-term volatility and large shocks. Forecasts are smoother than actual inflation, indicating information loss.

5.2 Economic Interpretation

The negative AR(1) coefficient (−0.638) reflects the well-documented mean-reverting nature of inflation. When inflation rises above trend (as occurred during 2021–2022), the model predicts a return toward the long-run average, consistent with both the ECB's policy responses and the fundamental dynamics of supply and demand. However, the magnitude of mean reversion captured by the model is insufficient to fully account for large shocks, particularly when external factors (energy prices, supply disruptions) drive sustained inflation.

The positive MA coefficient indicates that current inflation is also influenced by lagged shocks, suggesting that some inflation momentum persists month-to-month. This is visible in clusters of high or low inflation and reflects the time required for price changes to filter through the economy.

5.3 Limitations and Directions for Future Research

In this paper, the main limitations come from the open avenues for further work:

1. **Univariate specification**: The ARMA(1,1) ignores potentially important exogenous variables such as oil prices, exchange rates, output gaps, or labor market tightness. A Vector Autoregression (VAR) or structural model incorporating these factors would likely improve forecasts.

2. **Structural breaks**: The 2021–2023 inflation surge was largely unprecedented in the sample, driven by extraordinary shocks (pandemic disruptions, geopolitical tensions). Standard time-series models estimated on historical data struggle to predict regime shifts. Time-varying parameter models or models with structural breaks might better accommodate such episodes.

3. **Seasonal adjustment**: Although monthly inflation rates are less seasonal than the underlying price index, there may be residual seasonality worth modeling (e.g., through seasonal ARIMA—SARIMA)

4. **Real-time forecasting**: Our forecasts were generated with complete information through the estimation period. Real-time forecasts, using only data available at the time, would face additional data revisions and judgment calls about preliminary versus final inflation figures.

5. **Non-linear dynamics**: Inflation may exhibit threshold effects or asymmetries (e.g., faster pass-through of input cost shocks during elevated inflation). Non-linear models (regime-switching ARMA, Markov-switching models) could capture such behavior

6. **Comparison with other methods**: Benchmarking against survey forecasts (ECB SPF, Consensus Economics), machine learning methods, or more sophisticated econometric models would contextualize our ARMA results

5.4 Concluding Remarks

The ARMA(1,1) model shows that a fairly simple time-series setup can already pick up important features of German inflation. With a mean absolute forecast error of about 0.29 percentage points and roughly a one-third improvement over a naive benchmark, it gives a reasonable baseline for short-term inflation forecasts. At the same time, the model clearly has limits: it tends to underestimate volatility and cannot really predict big unexpected shocks. This suggests that univariate time-series tools are useful, but they should be combined with economic reasoning and richer, possibly multivariate, models.

For policymakers and practitioners, the main message is that inflation forecasting will always be uncertain. Models are better seen as devices to structure thinking than as exact prediction machines. In practice, combining the mean-reverting behaviour captured by the ARMA model with leading indicators and expert judgment would probably lead to more robust forecasts

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